

ShopKart Data Engineering Platform

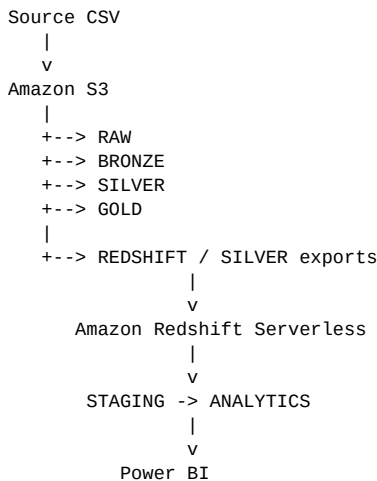
AWS Phase Documentation — Amazon S3 + Amazon Redshift Serverless

Purpose: GitHub-ready documentation of the AWS storage and warehouse layer used by the ShopKart data engineering project.

1. AWS Phase Overview

The AWS phase provides the cloud storage and analytics warehouse foundation for ShopKart. Amazon S3 is used for the data lake layers, while Amazon Redshift Serverless is used for the warehouse and star-schema analytics layer. The screenshots supplied for this documentation show the ShopKart S3 bucket, its processing folders, the Redshift Serverless namespace/workgroup, and the Redshift Query Editor.

Architecture



2. AWS Region and Redshift Serverless

Resource	Observed / Project Value
AWS Region	Asia Pacific (Hyderabad) — ap-south-2
S3 Bucket	shopkart-data-2026
Redshift Namespace	shopkart-namespace
Redshift Workgroup	shopkart-workgroup
Redshift Database	dev
Warehouse Type	Amazon Redshift Serverless

3. Amazon S3 Data Lake Structure

The supplied S3 screenshot shows the ShopKart bucket containing the following top-level folders:

```
shopkart-data-2026/
├── raw/
├── bronze/
├── silver/
├── gold/
├── redshift/
└── redshift_clean/
```

The **gold/** folder shown in the screenshot contains analytics outputs such as category_sales, customer_analytics, daily_sales, data_quality_results, overall_kpi, payment_analytics and product_performance.

4. S3 Gold Analytics Layer

```
gold/  
■■■ category_sales/  
■■■ customer_analytics/  
■■■ daily_sales/  
■■■ data_quality_results/  
■■■ overall_kpi/  
■■■ payment_analytics/  
■■■ product_performance/
```

5. Redshift Serverless Setup

The supplied AWS console screenshot shows both the default Redshift resources and the ShopKart resources. The ShopKart namespace and workgroup are displayed as Available.

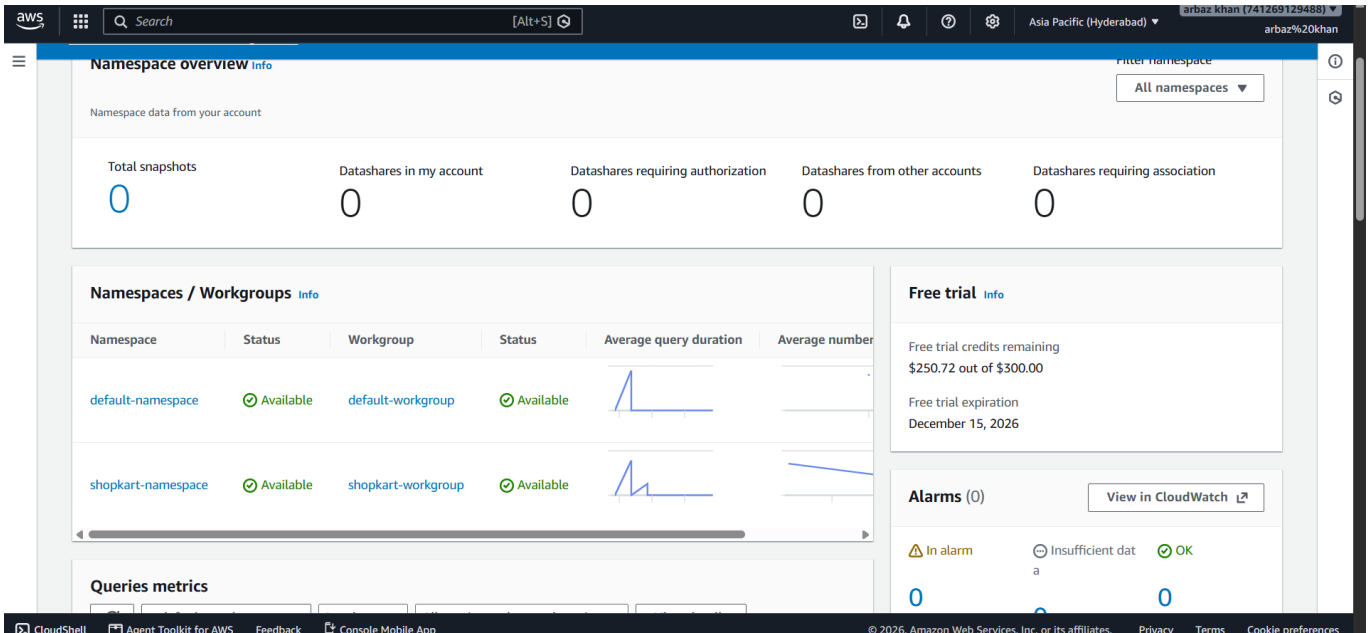


Figure 1 — Redshift Serverless namespace/workgroup view.

6. Redshift Query Editor

The Query Editor screenshot shows the ShopKart Redshift workgroup connection and SQL used to create the staging layer. The staging schema contains four tables: customers, products, orders and payments.

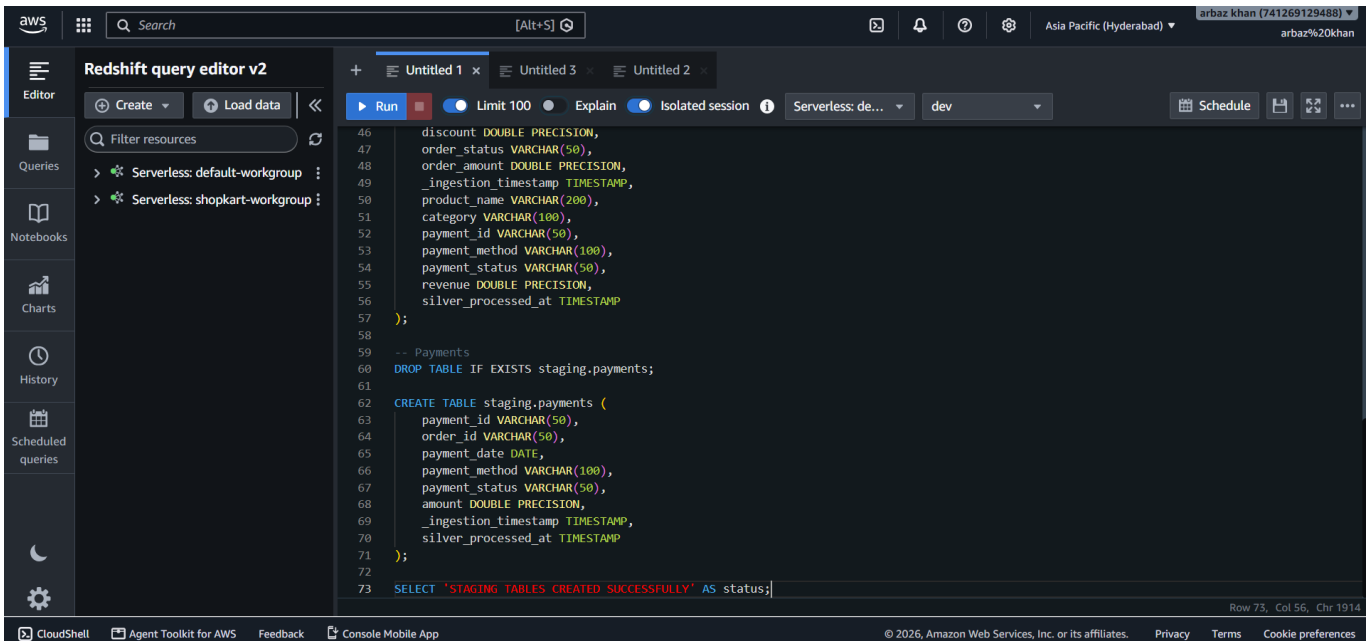


Figure 2 — Redshift Query Editor with ShopKart staging SQL.

7. Staging Schema

```
CREATE SCHEMA IF NOT EXISTS staging;
```

```
-- Customers
CREATE TABLE staging.customers (
  customer_id VARCHAR(50),
  customer_name VARCHAR(200),
```

```

    email VARCHAR(255),
    city VARCHAR(100),
    state VARCHAR(100),
    country VARCHAR(100),
    signup_date DATE,
    customer_segment VARCHAR(100),
    _ingestion_timestamp TIMESTAMP,
    silver_processed_at TIMESTAMP
);

-- Products
CREATE TABLE staging.products (
    product_id VARCHAR(50),
    product_name VARCHAR(200),
    category VARCHAR(100),
    subcategory VARCHAR(100),
    price DOUBLE PRECISION,
    cost DOUBLE PRECISION,
    _ingestion_timestamp TIMESTAMP,
    silver_processed_at TIMESTAMP
);

-- Orders
CREATE TABLE staging.orders (
    order_id VARCHAR(50),
    customer_id VARCHAR(50),
    product_id VARCHAR(50),
    order_date DATE,
    quantity INTEGER,
    discount DOUBLE PRECISION,
    order_status VARCHAR(50),
    order_amount DOUBLE PRECISION,
    _ingestion_timestamp TIMESTAMP,
    product_name VARCHAR(200),
    category VARCHAR(100),
    payment_id VARCHAR(50),
    payment_method VARCHAR(100),
    payment_status VARCHAR(50),
    revenue DOUBLE PRECISION,
    silver_processed_at TIMESTAMP
);

-- Payments
CREATE TABLE staging.payments (
    payment_id VARCHAR(50),
    order_id VARCHAR(50),
    payment_date DATE,
    payment_method VARCHAR(100),
    payment_status VARCHAR(50),
    amount DOUBLE PRECISION,
    _ingestion_timestamp TIMESTAMP,
    silver_processed_at TIMESTAMP
);

```

8. S3 Redshift Export Area

The supplied S3 screenshot shows a **redshift/** prefix containing exported analytics and Silver Parquet datasets. The visible folders include `category_sales`, `customer_analytics`, `daily_sales`, `overall_kpi`, `payment_analytics`, `product_performance`, `silver_customers`, `silver_orders`, `silver_payments` and `silver_products`.

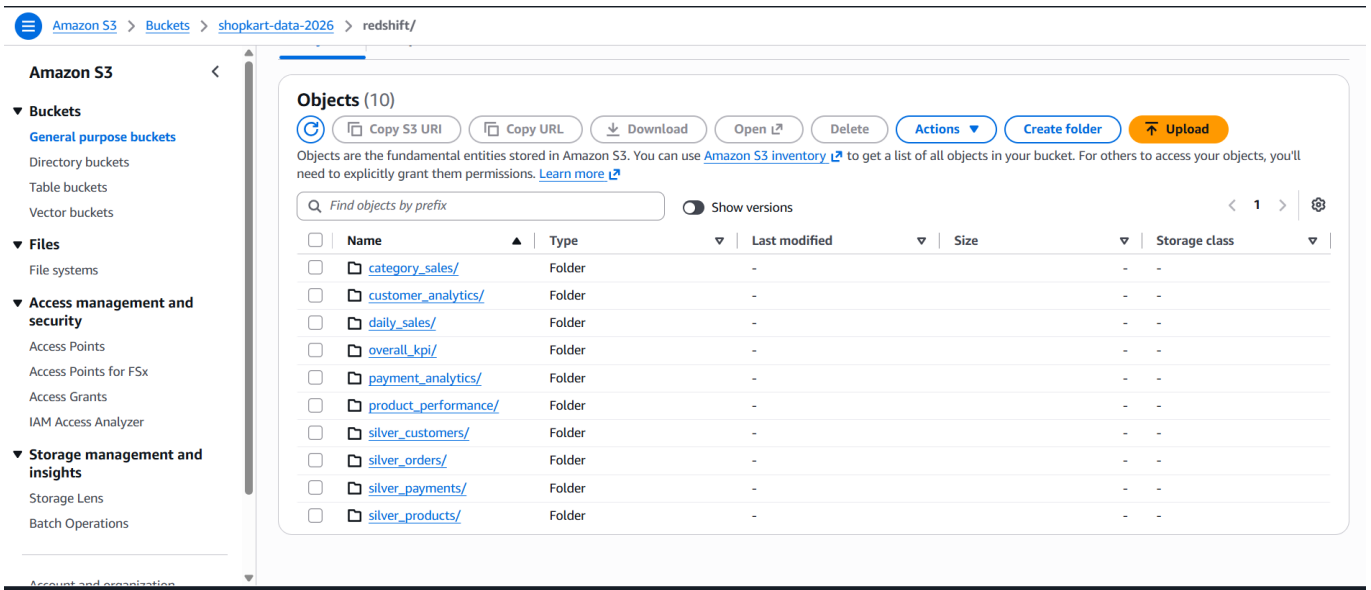


Figure 3 — S3 redshift/ prefix and Parquet export folders.

9. S3 Bucket Overview

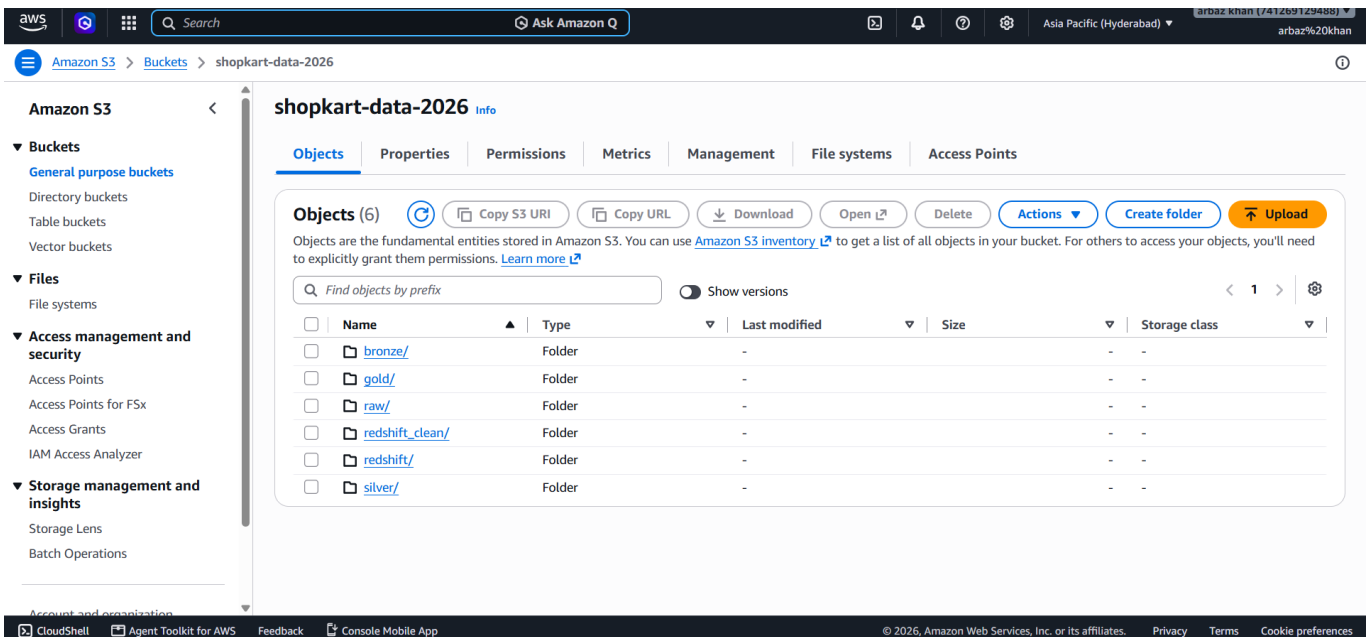


Figure 4 — ShopKart S3 bucket top-level data layers.

10. Gold Layer in S3

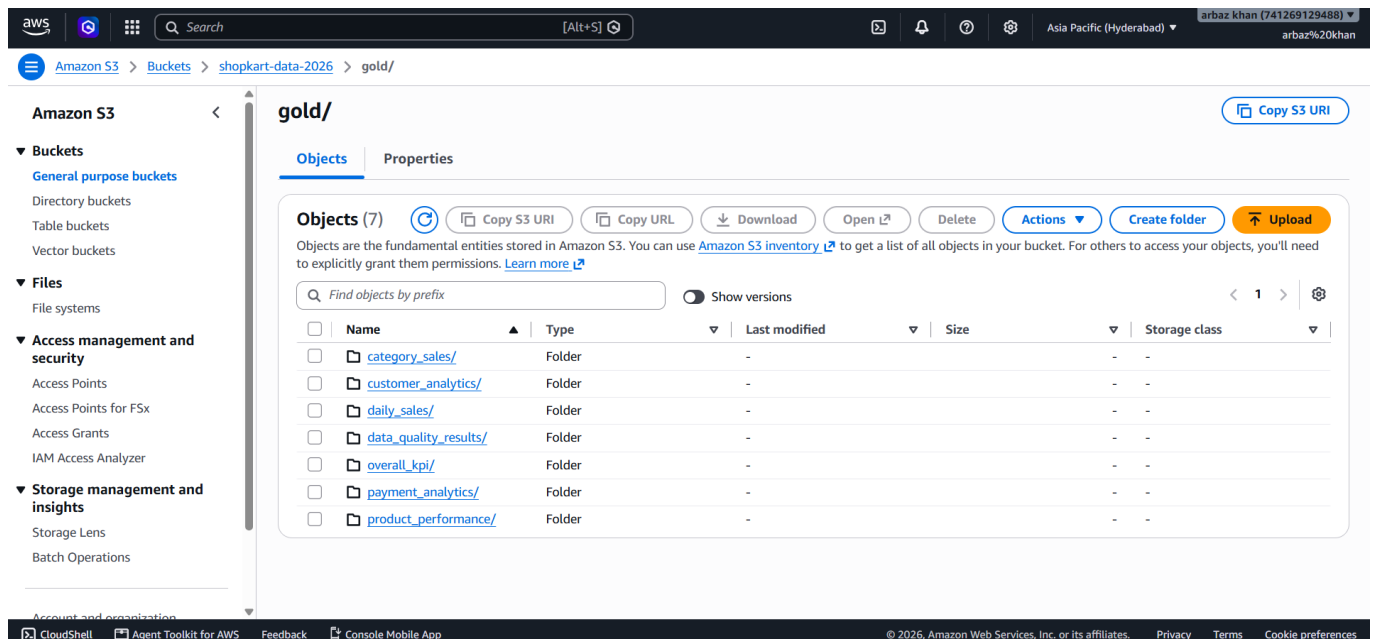


Figure 5 — ShopKart gold analytics folders in S3.

11. AWS → Redshift Data Flow

```
Silver Parquet files in S3
|
| COPY / Parquet load
V
Redshift STAGING
|
+--> staging.customers
+--> staging.products
+--> staging.orders
+--> staging.payments
|
V
Redshift ANALYTICS
|
+--> dim_customer
+--> dim_product
+--> dim_date
+--> dim_payment
+--> fact_sales
|
V
Power BI
```

12. GitHub Structure for AWS Phase

```
aws/
■■■■ README.md
■■■■ s3_structure.md
■■■■ iam_setup.md
■■■■ redshift_setup.md
■■■■ sql/
■ ■■■■ 01_staging_tables.sql
■ ■■■■ 02_dimension_tables.sql
■ ■■■■ 03_fact_sales.sql
■ ■■■■ 04_validation.sql
■■■■ screenshots/
■■■■ s3_bucket.png
■■■■ s3_gold.png
■■■■ s3_redshift_exports.png
■■■■ redshift_namespace.png
■■■■ redshift_query_editor.png
```

13. What Goes to GitHub vs AWS

GitHub	AWS Account
SQL scripts	S3 bucket
S3 structure documentation	Actual S3 objects
Redshift table definitions	Redshift Serverless workgroup
CLI/setup commands	IAM roles and permissions
Architecture diagrams	Running cloud resources
Screenshots	Actual data

14. Security — Never Upload Secrets

Never commit: AWS access keys, secret keys, session tokens, private keys, Redshift passwords, Databricks tokens, or .env files. Use IAM roles/connections and environment-specific secret management instead. Bucket names and SQL schema definitions can be documented when they do not expose credentials.

15. Git Commands

```
git status
git add aws/
```

```
git commit -m "Add AWS S3 and Redshift phase documentation"
git push
```

16. Interview Explanation

“For the ShopKart project, I used Amazon S3 as the cloud data lake with RAW, Bronze, Silver and Gold layers. The Silver and Gold outputs were stored in S3 as Parquet/Delta data. I then used Amazon Redshift Serverless as the analytical warehouse. Silver datasets were loaded into Redshift staging tables, and the warehouse layer was organized into dimensions and a fact table for Power BI reporting. I also worked with the Redshift Query Editor and IAM-based access between AWS services.”

Source basis: the AWS screenshots and SQL supplied in this conversation. This document intentionally records only what those materials support and does not expose credentials.